

Credit Card Fraud Detection using Machine Learning

Project Proposal

Machine Learning – End Semester Project

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Chapter 1

Problem Statement

Credit card fraud causes substantial financial losses worldwide, and the volume of transactions makes manual review impossible. The task is to build a system that classifies a transaction as fraudulent or legitimate from its attributes.

What makes this a demanding machine learning problem is not the classification itself but the **extreme rarity of the positive class**. In the benchmark dataset proposed for this project, fraudulent transactions account for approximately **0.17%** of all records. At that prevalence a classifier that ignores its input entirely and predicts “legitimate” for every transaction attains over **99.8% accuracy** while detecting no fraud whatsoever.

The project must therefore address two questions rather than one: which algorithm detects fraud best, and how should “best” be measured when the conventional metric is uninformative.

Chapter 2

Objectives

1. To apply the algorithms taught across **all five units** of the Machine Learning syllabus to a single real classification problem, rather than studying them in isolation.
2. To establish a single leakage-free experimental protocol so that every model is trained and evaluated on identical data and the comparison is valid.
3. To evaluate the models using metrics appropriate to extreme class imbalance – recall, precision, F1-score and PR-AUC – and identify the best performer.
4. To verify several theoretical results from the syllabus experimentally rather than merely restating them.
5. To produce a fully reproducible implementation, seeded with the last two digits of the author's roll number (77) as required.

Chapter 3

Scope

In scope. Supervised binary classification on the ULB benchmark dataset; the algorithms listed in the methodology below; feature selection and feature reduction; unsupervised clustering as a contrasting approach; evaluation by held-out test set and by cross validation.

Out of scope. Deep learning architectures beyond a multilayer perceptron; real-time deployment; datasets other than the chosen benchmark.

Chapter 4

Dataset

The Credit Card Fraud Detection dataset published by the Machine Learning Group of the Université Libre de Bruxelles.

Property	Value
Source	Kaggle (mlg-ulb/creditcardfraud)
Transactions	284,807
Attributes	30 (Time, V1-V28, Amount) plus the class label
Fraudulent	492 (0.172%)
Period	Two days, September 2013, European cardholders

Attributes V1 to V28 are principal components obtained by applying PCA to the original transaction attributes before publication, to protect cardholder confidentiality. Only Time, Amount and Class are untransformed. That the benchmark is itself the output of a Unit II technique is a point the project will examine directly.

Chapter 5

Methodology

5.1 Data preparation

Exact duplicate rows will be removed before any other processing, since a duplicate split across the training and test sets would cause the model to be evaluated on a record it has memorised. A single **stratified** 70/30 split with `random_state = 77` will then be made and used identically by every model. Standardisation will be performed inside pipelines so that no statistic computed on the test set influences training.

5.2 Algorithms, by syllabus unit

Unit	Topic	Planned implementation
I	Training, testing, validation; overfitting and underfitting	Stratified split, 5-fold cross validation, a model-complexity sweep
II	Decision tree learning, ID3	Decision tree with the entropy criterion; information gain computed by hand
II	Instance based learning	k-nearest neighbour with a worked numerical example

Unit	Topic	Planned implementation
II	Curse of dimensionality	Measured directly on the dataset
II	Feature selection	Univariate (F-test, mutual information), multivariate (RFE), embedded
II	Feature reduction	Principal Component Analysis and Linear Discriminant Analysis
III	Bayesian learning	Gaussian naive Bayes; the Bayes optimal decision rule derived from a loss matrix
III	Support Vector Machine	RBF-kernel SVM; dual formulation, soft margin and kernel comparison
IV	Artificial neural networks	McCulloch-Pitts neuron, perceptron learning logic gates, the XOR limitation, backpropagation
IV	Computational learning theory	VC dimension and PAC sample complexity applied to the available data
V	Ensembles	Bagging, Random Forest, AdaBoost, XGBoost
V	Rule fusion	Fixed rules (majority, mean, product, max, min) and a trained rule (stacking)

Unit	Topic	Planned implementation
V	Clustering	K-means and hierarchical clustering as unsupervised detection

5.3 Evaluation

Accuracy will be reported only once, alongside the score of the trivial classifier, to demonstrate why it cannot be used. Models will be compared on precision, recall, F1-score and PR-AUC on the fraud class, supported by confusion matrices expressed in absolute counts, and confirmed by 5-fold stratified cross validation.

Chapter 6

Expected Outcomes

1. A comparative evaluation of twelve models on an identical test set, identifying the best performer with supporting evidence.
2. Experimental confirmation of four theoretical results from the syllabus: the perceptron convergence theorem, the curse of dimensionality, the Bayes optimal decision rule, and the variance reduction produced by bagging.
3. A demonstration that accuracy is unusable at this class imbalance.
4. A reproducible repository containing ten executed notebooks, all generated figures and result tables, and a written report.

Chapter 7

Tools and Technologies

Python 3.11 with scikit-learn, XGBoost, pandas, NumPy, SciPy, Matplotlib and Jupyter. Version control and distribution through Git and GitHub.

Chapter 8

Plan of Work

Phase	Activity	Deliverable
1	Literature review; dataset acquisition and cleaning	Reviewed sources, cleaned dataset
2	Exploratory analysis; experimental protocol	Notebook 01
3	Feature selection and reduction	Notebook 02
4	Unit II models: decision tree, k-NN	Notebooks 03-04
5	Unit III models: naive Bayes, SVM	Notebooks 05-06
6	Unit IV models: perceptron, multilayer network	Notebook 07
7	Unit V models: ensembles, rule fusion, clustering	Notebooks 08-09
8	Comparison, cross validation, conclusions	Notebook 10
9	Report, presentation, repository	Report PDF, slides, GitHub repository

Chapter 9

References

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